



One step at a time: Miller (second from left) teaches the split step to members of the Bishop O'Dowd Boys High School tennis team in Oakland, California. SPECIAL ARRANGEMENT

results seemed extremely counterproductive. I created several other stop-action analyses of men and women making the kind of errors that Sebastian Korda (**below**) made, including ones where the problem was simply being late. I found that if the shot they were reacting to was challenging by virtue of its speed, players were handicapped by being airborne and landing their split step late.

Why is the split step so important in tennis?

There are three reasons:

- ❶ It allows a player to come into balance.
- ❷ It allows for bending the knees to produce a forceful move to the ball.
- ❸ It allows a player to make their first move towards the ball as soon as possible.

This third purpose is becoming more important as the game speeds up. Some men's pros now serve at 130 mph. These serves reach the baseline in 0.5 seconds. Fast serves by women reach the returner in 0.6

seconds. 80-mph passing shots travel from the baseline to a good net position in 0.44 seconds. A split step with a hop that is three inches high will touch down in about 0.24 seconds, and the after-landing knee bend duration is roughly 0.2 seconds, so the players' first move to the ball cannot begin until 0.44 seconds after they initiate their hop.

So, it is extremely important for players to time their split step so they touch down on the court early enough to complete their after-landing knee bend as soon as they detect the ball's direction — which can only be done with 100 per cent confidence after their opponent hits the ball. If they touch down later, they risk not being prepared for fast shots.

Who has executed the best split steps in tennis history?

Pete Sampras (the 1990s superstar) executed the best split steps. Many experts think Sampras had the best serve. I have not seen any mention of Sampras's serve-and-volley footwork, but close observation of that footwork is revealing and provides part of the explanation of why his serve is so highly rated.

Sampras, like all excellent servers, leapt when he served. He landed on his left foot, took one step with his right foot, then hopped into his split step. It was LAND-STEP-SPLIT. Very few other players do this. Most players do LAND-STEP-STEP-SPLIT or even LAND-STEP-STEP-STEP-SPLIT. Either of these options could be labelled as 'frolicking into the net'. They result in players being in the air or on one foot as they run, following the adage to 'get as close to the net as possible'.

Sampras was, instead, attempting to land his split step at the instant of his opponent's return, for instance against Andre Agassi, thereby giving himself time to take as many steps as

